Mat 2540 HW4

1. List all the permutations of $\{a, b, c\}$.

Sol: abc, acb, bac, bca, cab, cba

2. How many different permutations are there of the set $\{a, b, c, d, e, f, g\}$?

Sol: $P(7,7) = \frac{7!}{(7-7)!} = \frac{7!}{0!} = 7! = 5040$

3. How many permutations of $\{a, b, c, d, e, f, g\}$ end with a?

Sol. First we fix "a" to the end and then there are $P(6,6)$ permutations. $\Rightarrow P(6,6) = \frac{6!}{0!} = 6! = 720$

4. Let $S = \{1, 2, 3, 4, 5\}$.

a) List all the 3-permutations of S.
b) List all the 3-combinations of S.

Sol: a) 123 family: 123, 132, 213, 231, 312, 321
124 family: 124, 142, 214, 241, 412, 421
125 family: 125, 152, 215, 251, 512, 521
134 family: 134, 143, 314, 341, 413, 431
135 family: 135, 153, 315, 351, 513, 531
145 family: 145, 154, 415, 451, 514, 541
234 family: 234, 243, 324, 342, 423, 432
235 family: 235, 253, 325, 352, 523, 532
245 family: 245, 254, 425, 452, 524, 542
345 family: 345, 354, 435, 453, 534, 543

b) $\{1,2,3\}$, $\{1,2,4\}$, $\{1,2,5\}$, $\{1,3,4\}$, $\{1,3,5\}$
$\{1,4,5\}$, $\{2,3,4\}$, $\{2,3,5\}$, $\{2,4,5\}$, $\{3,4,5\}$

5. Find the value of each of these quantities.

a) $P(6, 3)$ **b)** $P(6, 5)$
c) $P(8, 1)$ **d)** $P(8, 5)$
e) $P(8, 8)$ **f)** $P(10, 9)$

a) $P(6,3) = \frac{6!}{(6-3)!} = \frac{6!}{3!} = \frac{6\times5\times4\times3\times2\times1}{3\times2\times1} = 6\times5\times4 = 120.$

b) $P(6,5) = \frac{6!}{(6-5)!} = \frac{6!}{1!} = 6! = 720$

c) $P(8,1) = \frac{8!}{(8-1)!} = \frac{8!}{7!} = 8$

d) $P(8,5) = \frac{8!}{(8-5)!} = \frac{8!}{5!} = 8\times7\times6 = 336$

e) $P(8,8) = \frac{8!}{(8-8)!} = \frac{8!}{0!} = 8! = 40320$

f) $P(10,9) = \frac{10!}{(10-9)!} = \frac{10!}{1!} = 10!$

6. Find the value of each of these quantities.

a) $C(5, 1)$ **b)** $C(5, 3)$
c) $C(8, 4)$ **d)** $C(8, 8)$
e) $C(8, 0)$ **f)** $C(12, 6)$

a) $C(5,1) = \frac{5!}{1!(5-1)!} = \frac{5!}{1!4!} = \frac{5}{1!} = 5$

b) $C(5,3) = \frac{5!}{3!(5-3)!} = \frac{5!}{3!2!} = \frac{5\times4\times3}{3!} = \frac{5\times4\times3}{3\times2\times1} = 10$

c) $C(8,4) = \frac{8!}{4!4!} = \frac{8\cdot7\cdot6\cdot5}{4\cdot3\cdot2\cdot1} = 70$

d) $C(8,8) = \frac{8!}{8!0!} = 1$

e) $C(8,0) = \frac{8!}{0!8!} = 1$

f) $C(12,6) = \frac{12!}{6!6!} = \frac{\overset{2}{\cancel{12}}\times11\times\overset{2}{\cancel{10}}\times\overset{3}{\cancel{9}}\times\overset{\cancel{2}}{\cancel{8}}\times7}{\cancel{6}\times\cancel{5}\times\cancel{4}\times\cancel{3}\times\cancel{2}\times1} = 12\times11\times7 = 132\times7 = 924$

7. Find the number of 5-permutations of a set with nine elements.

$P(9,5) = \frac{9!}{(9-5)!} = \frac{9!}{4!}$

$= 9\times8\times7\times6\times5 = 15120$

9. How many possibilities are there for the win, place, and show (first, second, and third) positions in a horse race with 12 horses if all orders of finish are possible?

This equals to 3-permutations of 12 elements:

$P(12,3) = \frac{12!}{(12-3)!} = \frac{12!}{9!} = 12\times11\times10 = 1320.$

11. How many bit strings of length 10 contain

a) exactly four 1s?
b) at most four 1s?
c) at least four 1s?
d) an equal number of 0s and 1s?

a) exactly four 1s

$\Rightarrow$ pick 4 digits out of 10 and put 1s in it.

$\Rightarrow C(10,4) = \frac{10!}{4!\,6!} = \frac{10\cdot 9\cdot 8\cdot 7}{4!} = 210$

b) "at most four 1s" equals

"exactly four 1s" $\cup$ "exactly three 1s" $\cup$ "exactly two 1s" $\cup$ "exactly one 1" $\cup$ "NO 1" and

we have $C(10,4) + C(10,3) + C(10,2) + C(10,1) + C(10,0)$

$$= 210 + \frac{10!}{3!\,7!} + \frac{10!}{2!\,8!} + \frac{10!}{1!\,9!} + \frac{10!}{1!\,10!}$$

$$= 210 + \frac{10\cdot \cancel{9}^{3}\cdot \cancel{8}^{4}}{\cancel{3}\cdot\cancel{2}\cdot 1} + \frac{10\cdot 9}{2\cdot 1} + \frac{10}{1} + 1$$

$$= 210 + 120 + 45 + 10 + 1 = 386$$

c) "at least four 1s" equals

"exactly four 1s" $\cup$ "exactly five 1s" $\cup$ "exactly six 1s" $\cup$ "exactly seven 1s" $\cup$ "exactly eight 1s" $\cup$ "exactly nine 1's" $\cup$ "exactly ten 1s", and we have.

$$C(10,4) + C(10,5) + C(10,6) + C(10,7) + C(10,8) + C(10,9) + C(10,10)$$

$$= 210 + \frac{10!}{5!\,5!} + 210 + 120 + 45 + 10 + 1$$

$$= 210 + \frac{\cancel{10}^{2}\cdot 9\cdot \cancel{8}^{2}\cdot \cancel{7}\cdot 6}{\cancel{5}\cdot\cancel{4}\cdot\cancel{3}\cdot\cancel{2}\cdot 1} + 386 = 210 + 252 + 386 = 848$$

d) an equal number of 0s and 1s:

It means five 1s and five 0s. $\Rightarrow$ exactly five 1's. (or exactly five 0's).

$$C(10,5) = \frac{10!}{5!\,5!} = 252$$

17. How many subsets with more than two elements does a set with 100 elements have?

Sol: "Number of subsets with more than two elements from 100 elements" equals

"Total number of subsets" minus "number of subsets with less than or equal to two elements"

$|S| = 100$ (total number of elements)

"Total number of subsets" $= 2^{|S|} = 2^{100}$.

"number of subsets with less than or equal to two elements" $= C(100,0) + C(100,1) + C(100,2)$

empty set — subset with 1 element — subset with 2 elements

$$= 1 + \frac{100!}{1!\,99!} + \frac{100!}{2!\,98!}$$

$$= 1 + 100 + 4950 = 5051$$

$\Rightarrow$ Number of subsets with more than two elements from 100 elements

$$= 2^{100} - 5051.$$

19. A coin is flipped 10 times where each flip comes up either heads or tails. How many possible outcomes

a) are there in total?
b) contain exactly two heads?
c) contain at most three tails?
d) contain the same number of heads and tails?

Sol a) only 2 outcome from each flip (head or tail)

If one flips 10 times, the total number of outcome is $2^{10} = 1024$

b) "contain exactly two heads" = pick 2 "positions" out of 10:

$$C(10,2) = \frac{10!}{2!\,8!} = \frac{\overset{5}{\cancel{10}}\cdot 9}{\cancel{2}} = 45$$

c) "contain at most three tails" equals

"contain 0 tail" $\cup$ "contain 1 tail" $\cup$ "contain 2 tails" $\cup$ "contain 3 tails"

$$C(10,0) + C(10,1) + C(10,2) + C(10,3)$$

$$= \frac{10!}{0!\,10!} + \frac{10!}{1!\,9!} + \frac{10!}{2!\,8!} + \frac{10!}{3!\,8!}$$

$$= 1 + 10 + 45 + 120 = 176$$

d) "contain some number of heads and tails" = "contain 5 tails"

$$C(10,5) = \frac{10!}{5!\,5!} = \frac{10\cdot 9\cdot 8\cdot 7\cdot 6}{5\cdot 4\cdot 3\cdot 2\cdot 1} = 252$$

33. The English alphabet contains 21 consonants and five vowels. How many strings of six lowercase letters of the English alphabet contain

a) exactly one vowel?
b) exactly two vowels?
c) at least one vowel?
d) at least two vowels?

Sol. a) eactly one vowel for six letters : one vowel with 5 consonants

pick one vowel , pick a spot out of six for this vowel , the other 5 spots for consonants

$C(5,1) = \frac{5!}{1!4!} = 5$ ways

$C(6,1) = \frac{6!}{1!5!} = 6$

$21 \cdot 21 \cdot 21 \cdot 21 \cdot 21 = 21^5$

Total is $5 \cdot 6 \cdot 21^5 = 122,523,030.$

b) eactly two vowel for six letters: 2 vowels with 4 consonants.

pick 2 spots out of six for vowel. put 2 vowel in the spots put the other 4 consonant

$C(6,2) = \frac{6!}{2!4!} = 15$

5^2

21^4

Total is $15 \cdot 5^2 \cdot 21^4 = 72930375$

c) at least one vowel = Total number − No vowel

$= 26^6 - 21^6 = 223,149,655$

d) at least 2 vowel = Total number − No vowel − exactly 1 vowel

$= 26^6 - 21^6 -$ part a)

$= 100,626,625$